

ML Models Comparison on Iris Dataset

This notebook compares SVM, Logistic Regression, and KNN using Accuracy and Recall, along with visualization.

1. Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.metrics import accuracy_score, recall_score

from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
```

2. Load Dataset

```
df = pd.read_csv("Iris.csv")
df.head()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	
Species						
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

3. Prepare Data

```
X = df.drop(columns=["Species", "Id"])
y = df["Species"]

le = LabelEncoder()
y = le.fit_transform(y)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
```

```
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

4. Train Models

```
models = {
    "SVM": SVC(kernel='linear'),
    "Logistic Regression": LogisticRegression(),
    "KNN": KNeighborsClassifier(n_neighbors=5)
}

results = {}

for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)

    acc = accuracy_score(y_test, y_pred)
    rec = recall_score(y_test, y_pred, average='macro')

    results[name] = {"Accuracy": acc, "Recall": rec}

results

{'SVM': {'Accuracy': 0.9666666666666667, 'Recall': 0.9629629629629629},
 'Logistic Regression': {'Accuracy': 1.0, 'Recall': 1.0},
 'KNN': {'Accuracy': 1.0, 'Recall': 1.0}}
```

5. Convert Results to DataFrame

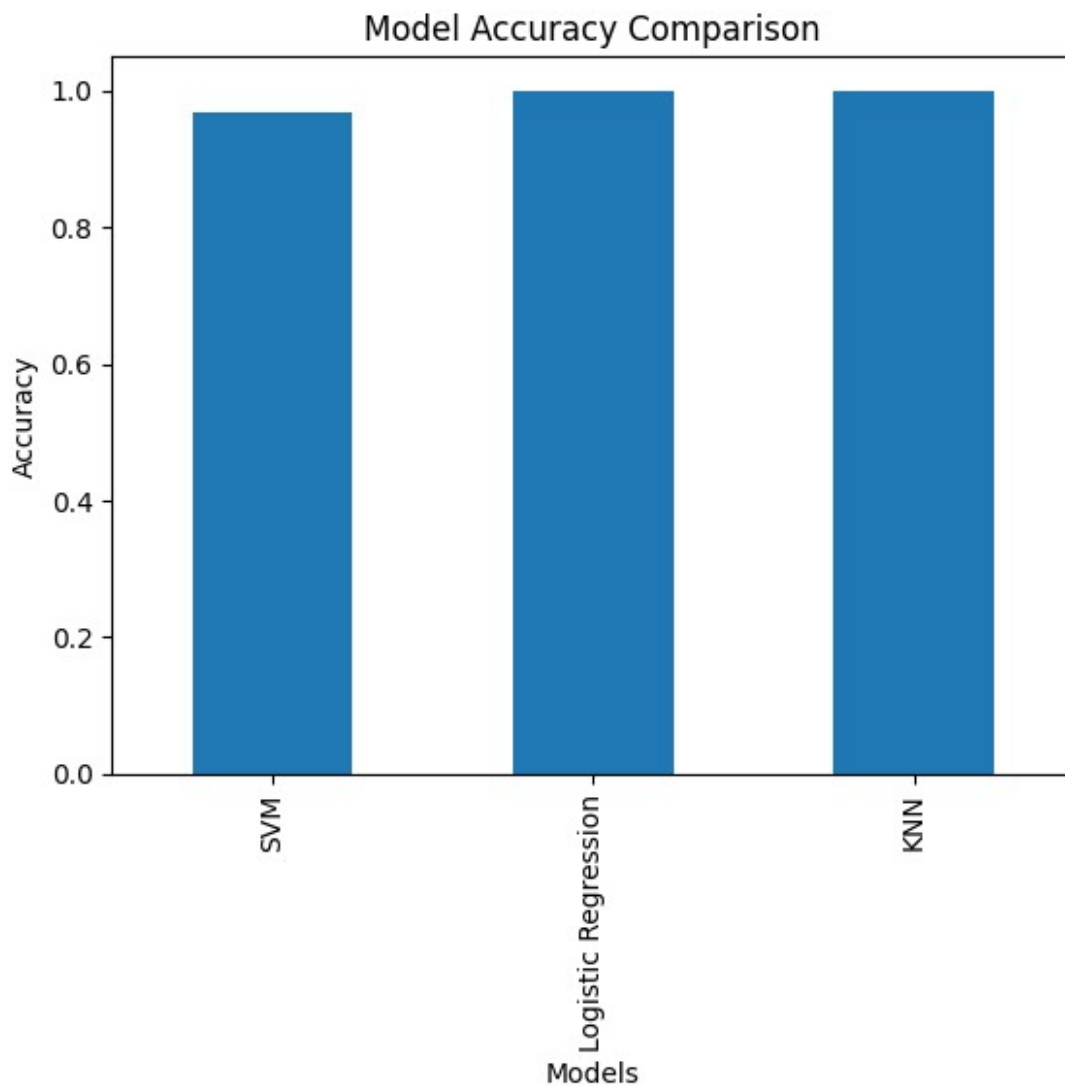
```
results_df = pd.DataFrame(results).T
results_df
```

	Accuracy	Recall
SVM	0.966667	0.962963
Logistic Regression	1.000000	1.000000
KNN	1.000000	1.000000

6. Plot Accuracy

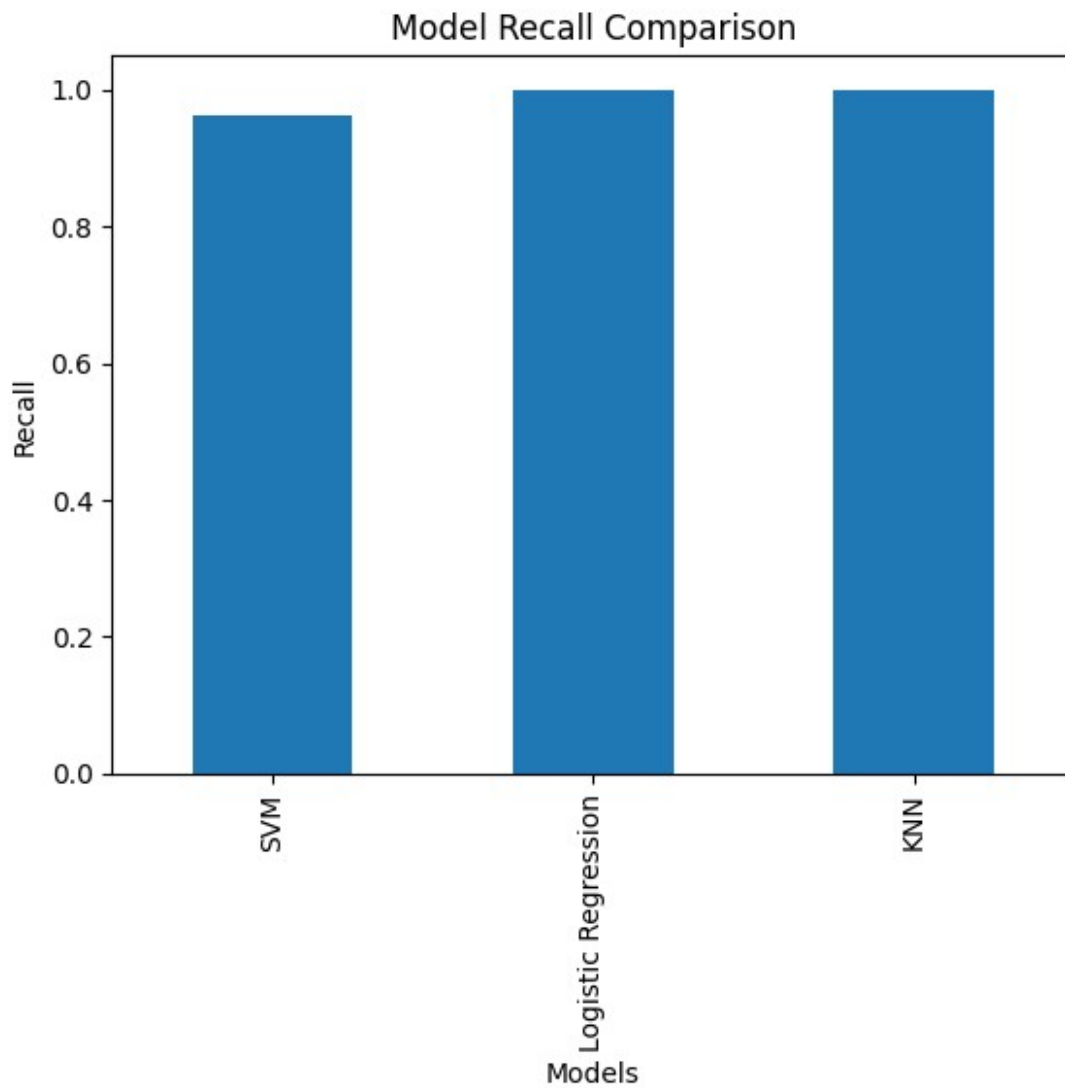
```
plt.figure()
results_df["Accuracy"].plot(kind='bar')
plt.title("Model Accuracy Comparison")
plt.xlabel("Models")
```

```
plt.ylabel("Accuracy")  
plt.show()
```



7. Plot Recall

```
plt.figure()  
results_df["Recall"].plot(kind='bar')  
plt.title("Model Recall Comparison")  
plt.xlabel("Models")  
plt.ylabel("Recall")  
plt.show()
```



Conclusion

This notebook compares three ML models:

- SVM
- Logistic Regression
- KNN

Metrics used: Accuracy and Recall.